# CGB1221 – DATABASE MANAGEMENT SYSTEMS
# END SEMESTER REVIEW

## Artificial Intelligence
## Academic Year: 2024 – 2025 (Even Semester)

| | | |
|---|---|---|
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| **Name** | **:** | **ROOBA.B** |
| **Year** | **:** | **II** |
| **Semester** | **:** | **IV** |
| **Section** | **:** | **B** |
| **Date** | **:** | **02.06.2025** |

# Title of the Project

## JOB PORTAL SYSTEM

# Abstract

A Job Portal System is a web-based application designed to streamline the job search and recruitment process using a Database Management System (DBMS). The system facilitates job seekers in searching and applying for relevant job opportunities while enabling employers to post vacancies and manage applications efficiently. The DBMS stores and organizes job listings, candidate profiles, resumes, and application statuses, ensuring easy retrieval and management of data. For job seekers, the platform provides a searchable job database, profile creation, and application tracking, enhancing their job-hunting experience. The system ensures data integrity, security, and efficient query processing, offering a seamless experience for both candidates and recruiters.

# Abstract with CO/PO Mapping

| Abstract | CO | POs | PSO |
|---|---|---|---|
| A Job Portal System is a web-based application designed to streamline the job search and recruitment process using a Database Management System (DBMS). The system facilitates job seekers in searching and applying for relevant job opportunities while enabling employers to post vacancies and manage applications efficiently. The DBMS stores and organizes job listings, candidate profiles, resumes, and application statuses, ensuring easy retrieval and management of data. For job seekers, the platform provides a searchable job database, profile creation, and application tracking, enhancing their job-hunting experience. The system ensures data integrity, security, and efficient query processing, offering a seamless experience for both candidates and recruiters. | CO1<br>CO2<br>CO3 | PO 1<br>PO 2<br>PO 3<br>PO 5<br>PO 6<br>PO 7<br>PO 8<br>PO 9<br>PO 10<br>PO 11<br>PO 12 | PSO1<br>PSO2 |

# Introduction

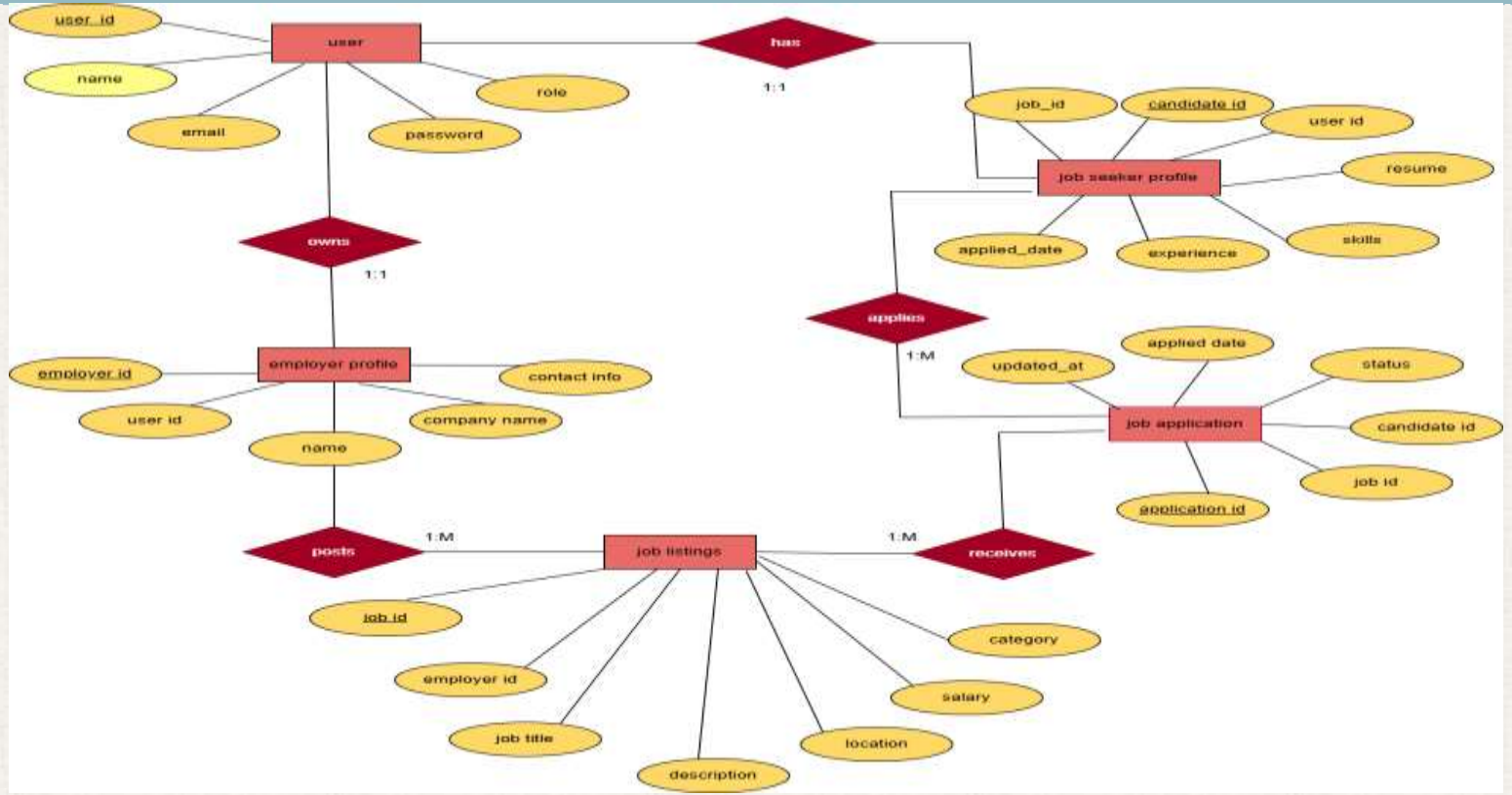
In today's digital era, job portals have become an essential bridge between job seekers and employers , revolutionizing the recruitment process. A **Job Portal System** is a web-based application designed to simplify and optimize job searching and hiring through the efficient use of a **Database Management System (DBMS).**

By leveraging a well-structured database, the system enables job seekers to explore relevant opportunities, create profiles, and track applications while allowing employers to post vacancies, filter candidates, and manage recruitment seamlessly. With an intuitive interface and robust data management capabilities, the Job Portal System enhances user experience by ensuring **efficient query processing, data security, and integrity**. The platform provides a centralized database for storing job listings, candidate details, resumes, and application statuses, making it easy to access and retrieve essential information.

# DBMS - Concepts Used

- Entity-Relationship (ER) Model
- Relational Database Concepts
- Transactions & Data Integrity
- Authentication and Authorization
- Query Optimization

# E-R DIAGRAM


user_id
user
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job seeker profile
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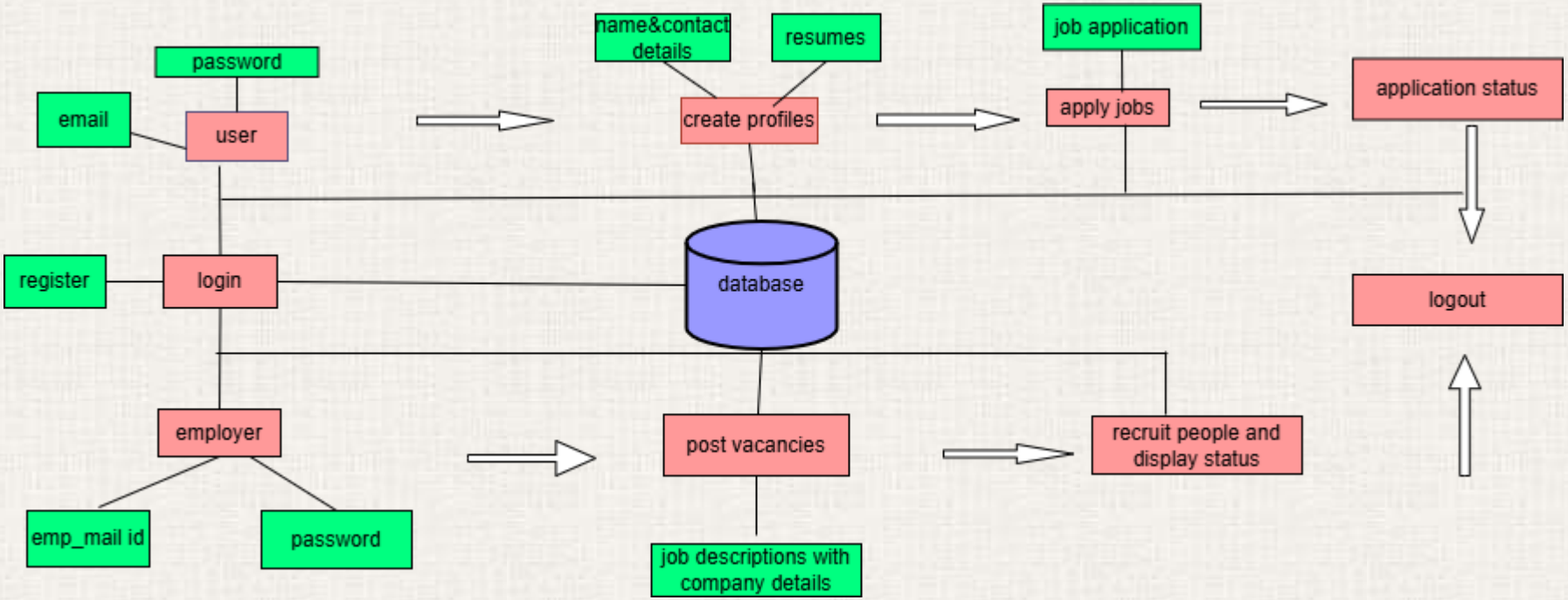
salary
job title
location
description

# Proposed Architecture



# Proposed Architecture - Description

The **Job Portal System** is designed to facilitate seamless interaction between **job seekers** and **employers**, ensuring efficient job search and hiring processes. At its core, the system maintains a **centralized database** that stores user information, job listings, and job applications. Users must first **register and authenticate**, where they can either be **job seekers** or **employers**. Job seekers can create a **candidate profile** that includes their resume, skills, and experience, allowing them to apply for job listings. Employers, on the other hand, can create an **employer profile**, which enables them to post job vacancies, specifying job titles, descriptions, salaries, and locations. The system also manages job applications, linking candidates to job postings and tracking application statuses as **pending, accepted, or rejected**. By structuring data through **strong and weak entities**, such as User, Job Listings, Job Seeker Profile, Employer Profile, and Job Application, the system efficiently organizes information, ensuring that all relationships between users, employers, and job opportunities are well-maintained. This model enhances the hiring experience by providing a structured, transparent, and user-friendly.

# List of Modules

- Job Seeker Module
- Employer Module
- Application Status Module

# Module Description

## 1.Job Seeker Module:

The **Job Seeker** module enables users to register and log in to the job portal system to actively participate in the job search process. Once logged in, job seekers can browse available job listings and apply for jobs that match their interests and qualifications. The job application includes essential information such as the candidate's resume, key skills, and work experience. After applying, job seekers can track the status of their applications in real time, with updates indicating whether the application is **Accepted**, **Rejected**, or still **Pending**. This module ensures a streamlined and user-friendly experience for candidates navigating their career opportunities through the portal.

# Module Description (Cont..)

## 2.Employer Module:

The **Employer** module allows company representatives to register and log in to the job portal system to manage recruitment activities for their respective organizations. Once authenticated, employers can post job openings with relevant details under their company's name. They can view applications submitted by job seekers for each job listing, including candidate resumes, skills, and experience. Employers have the ability to review and update the application status by accepting, rejecting, or marking applications as pending. This module provides employers with an efficient platform to manage job postings and streamline their hiring process.

# Module Description (Cont..)

## 3. Application Status Module:

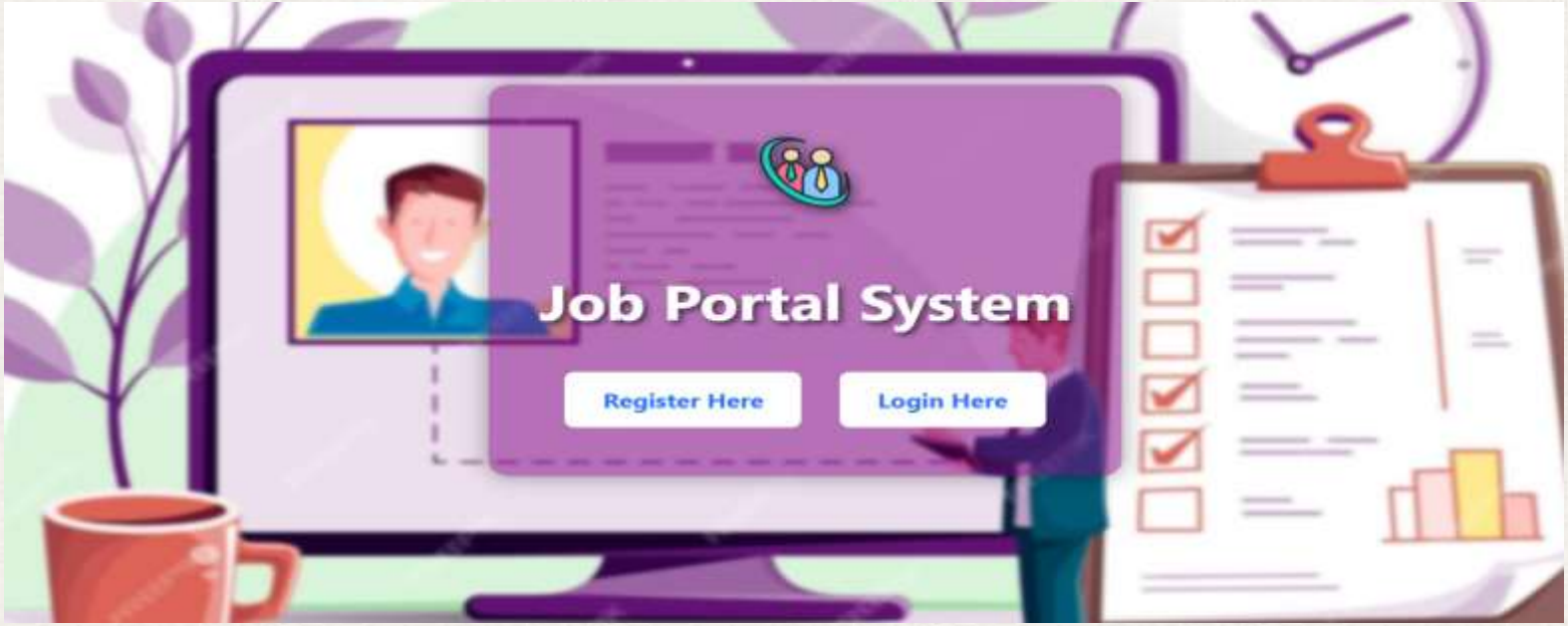
The **Application Status** module serves as the central interface for managing and tracking job applications submitted through the portal. It bridges the interaction between job seekers and employers by displaying the real-time status of each job application. Job seekers can view the current status of their applications—whether **Accepted**, **Rejected**, or **Pending**—for every job they have applied to. Simultaneously, employers can access and manage the list of applicants for each job posting, review their profiles, and update the status of each application accordingly. This module ensures transparency and seamless communication throughout the recruitment process.

# Software Specification
# (Tools / Programming Languages used)

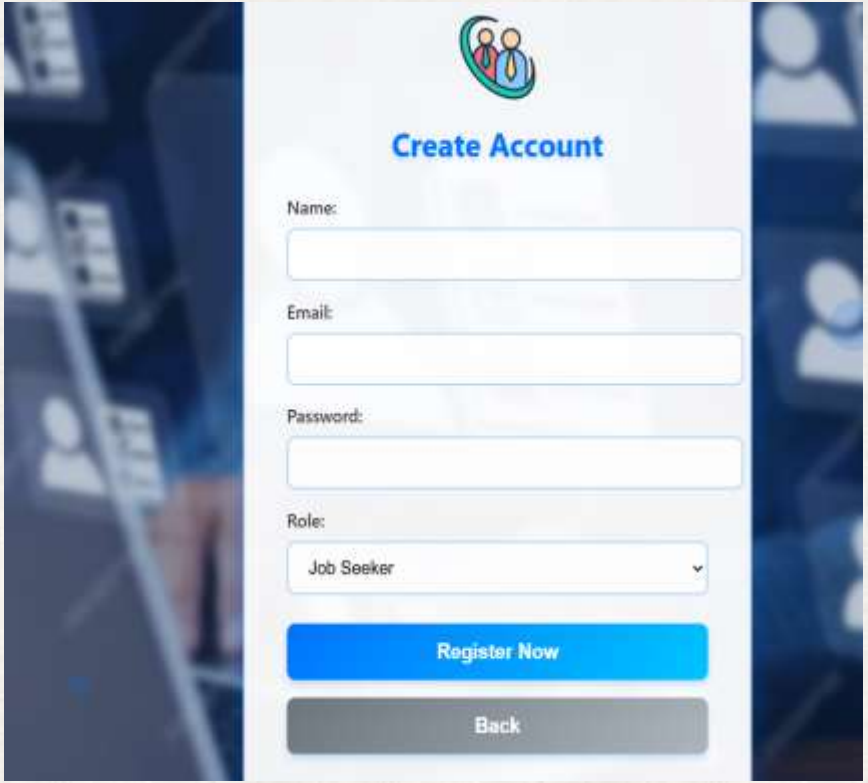
- Front-end:  HTML and CSS.

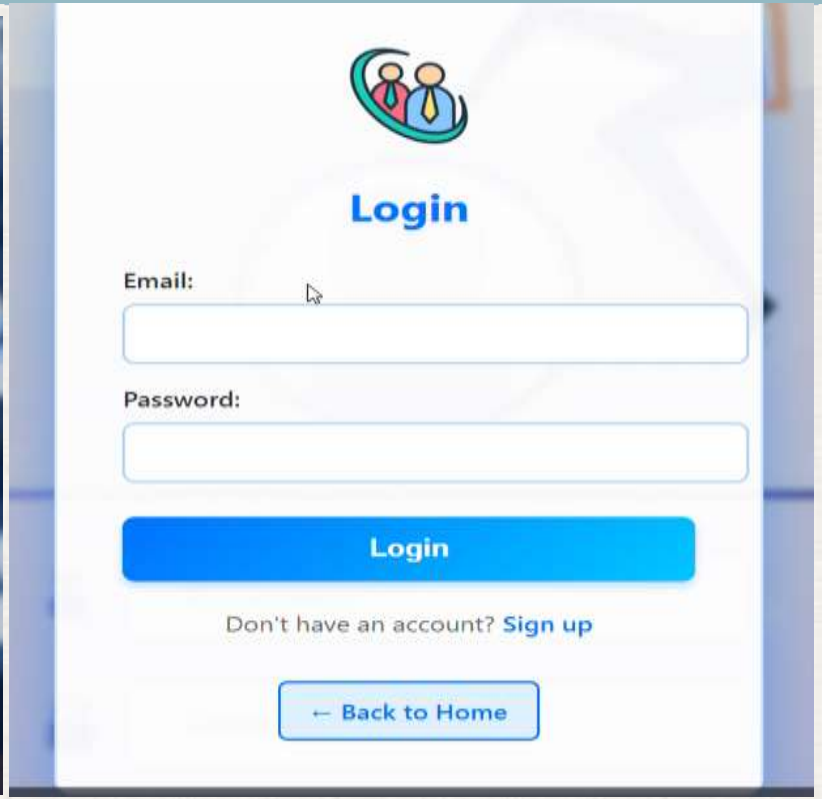
- Back-end: PHP and SQL(MYSQL).

# Job Portal System Screen shot



# Register / Login Screen shot



# Job Seeker Dashboard Screen shot

View Applications | Filter Job

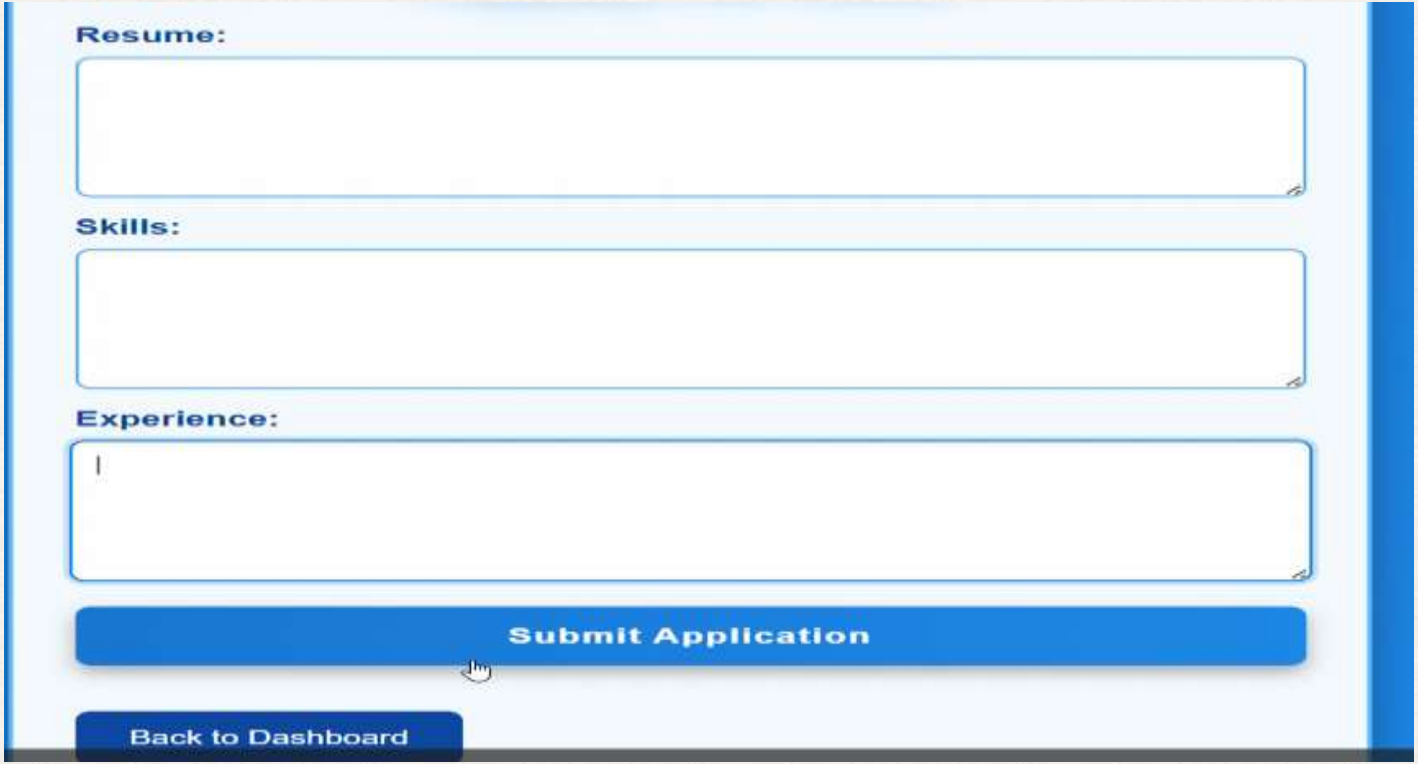
**Company:** Tata Consultancy Services (TCS)
**Job Title:** graphic designer
**Description:** knowledge in adobe photoshop, adobe illustrator, figma, adobe indesign and with 1 year of experience
**Location:** chennai
**Salary:** $30000
**Category:** designer

Apply Job

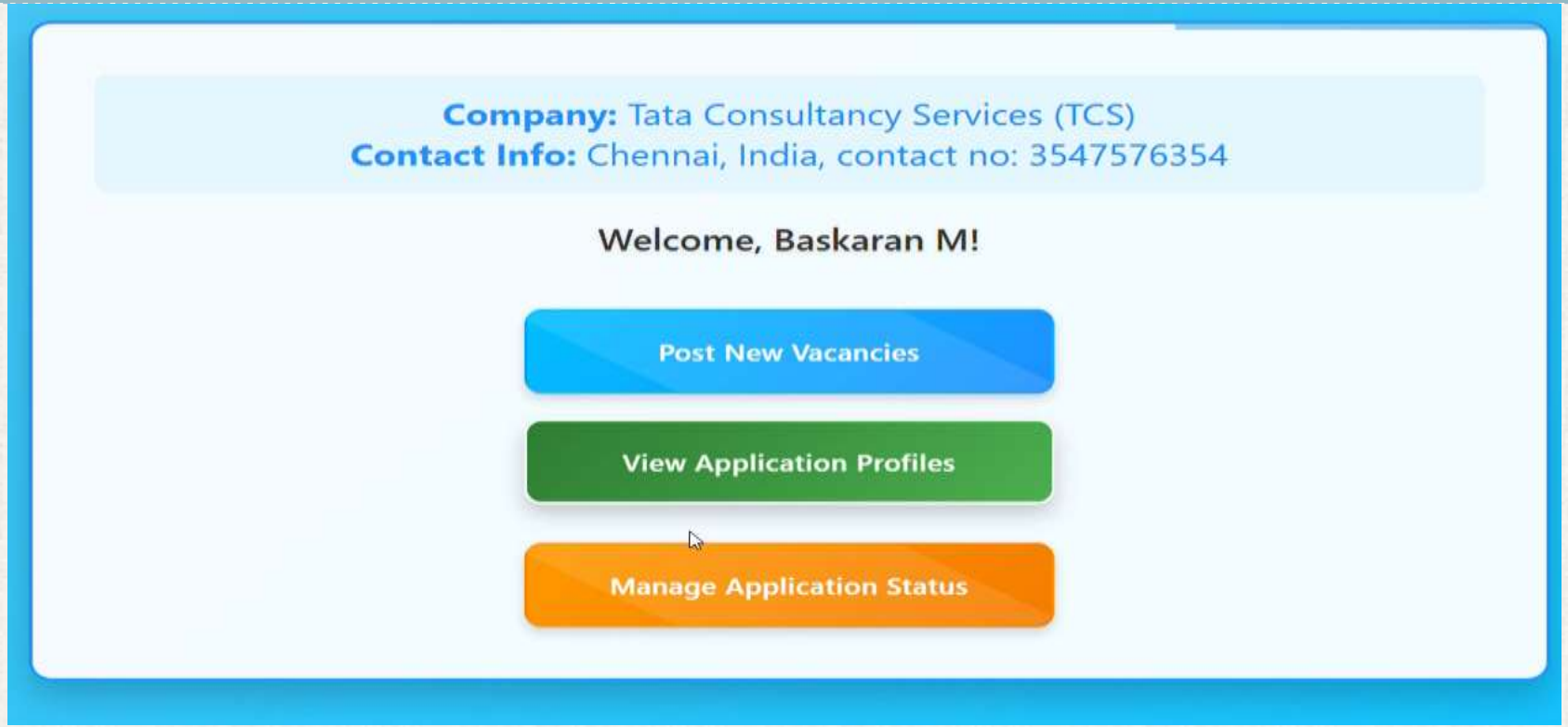
**Company:** Tata Consultancy Services (TCS)
**Job Title:** web development
**Description:** Frontend: A web developer should know HTML to structure web pages properly. Backend: Understanding PHP helps in building dynamic and interactive websites. Database: Knowledge of MySQL is essential for storing and retrieving data.
**Location:** chennai
**Salary:** $400000
**Category:** developer

Apply Job

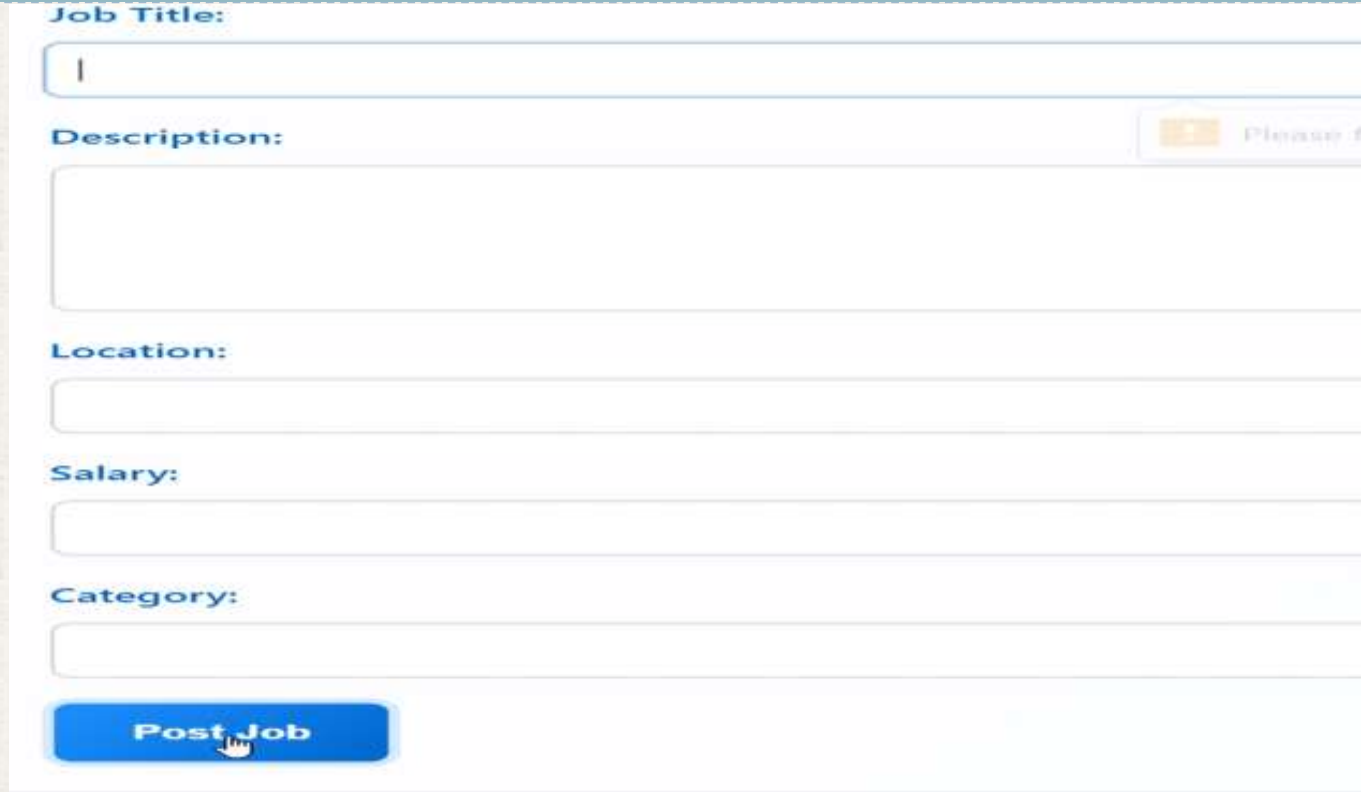
# Applying for a Job Screen shot



# Employer Dashboard Screen shot



# Posting a Job Screen shot



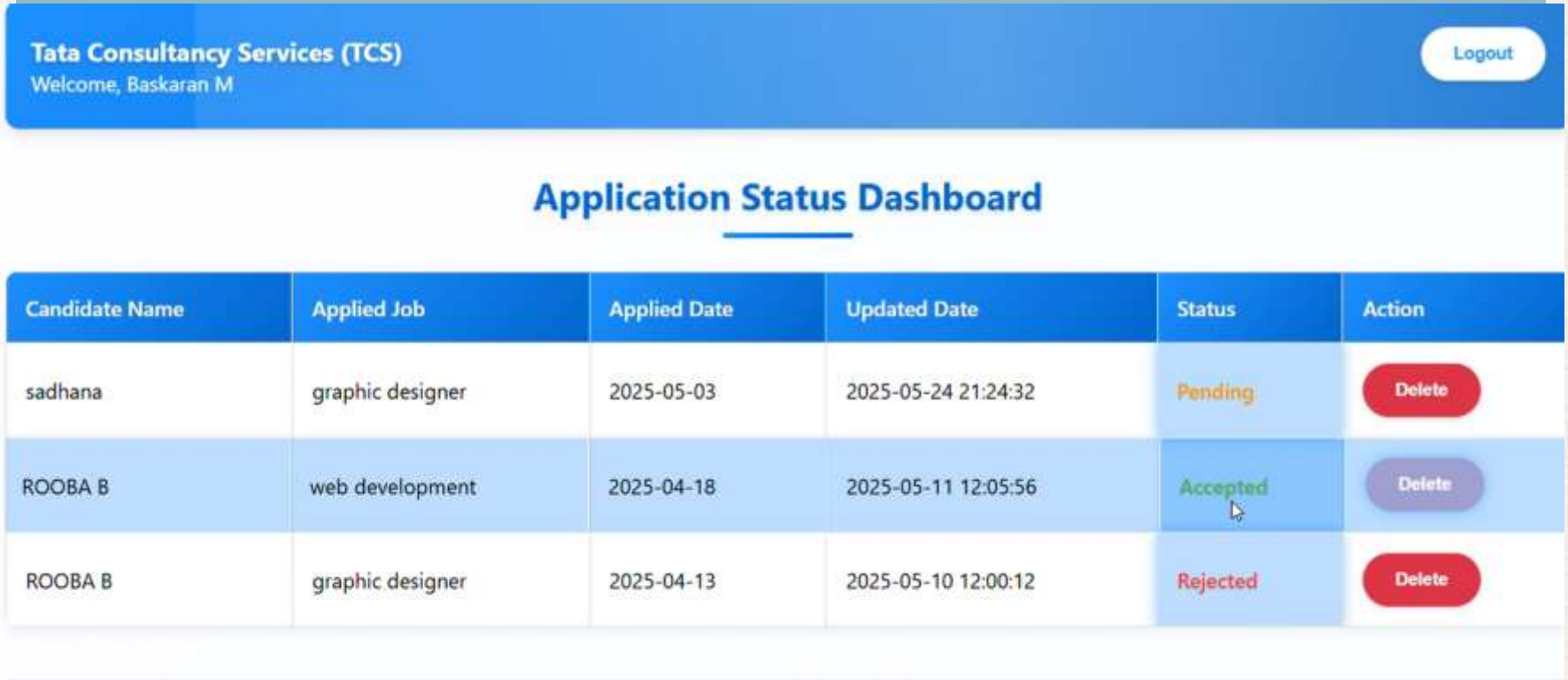
# Application Profiles Screen shot

## Application Profiles - Tata Consultancy Services (TCS)

Back to Dashboard

| Candidate Name | Job Title | Resume | Skills | Experience | Applied Date | Status |
|---|---|---|---|---|---|---|
| sadhana | graphic designer | Completed high school with a focus on science, achieving strong academic performance. Graduated with a Bachelor's degree in Computer Science from Anna University , specializing in software development. | Proficient in Adobe Creative Suite (Photoshop, Illustrator, InDesign) and UI/UX design principles. Skilled in creating visually engaging graphics, branding, and digital media for diverse projects. | 3 years | 2025-05-03 | Accepted Update |
| ROOBA B | web development | Completed schooling from Sri sankara vidhyalaiya, achieving strong academic results. Graduated from Annai magalir college with a degree in information technology(IT)/B.Tech Skilled in HTML, CSS, JavaScript, PHP, and MySQL with hands-on web development ex | Frontend: HTML, CSS, JavaScript, React, Vue.js, Backend: PHP, Node.js, Python, Ruby, Databases: MySQL, MongoDB, PostgreSQL Version Control: Git, GitHub, API: REST, GraphQL | 2 years | 2025-04-18 | Accepted Update |
| ROOBA B | graphic designer | i completed my schooling in meenakshi matric school and college in kumarasamy in karur and i am working as a data scientist in | i have experience dealing with powerbi,ai tools and scientific analysis tools and ai basic toools | 1 year | 2025-04-13 | Rejected Update |

# Application Status Screen shot



# Conclusion

The Job Portal System developed in this project is a dynamic web-based application designed to streamline the interaction between job seekers and employers. The **frontend** of the system is created using **HTML** and **CSS**, offering user-friendly interfaces for logging in, applying to jobs, and managing employer/job seeker dashboards. The **backend** is built with **PHP**, which handles the core logic such as session control, form handling, and server-side validations. All data storage and retrieval operations are performed using **MySQL**, a relational database system, with SQL queries managing tables like users, job_listings, employerprofile, jobseeker_profile, and jobapplication. The system supports role-based access, allowing employers to post jobs and view applications, while job seekers can apply for jobs and track their application status. The use of secure login with hashed passwords enhances data integrity and user privacy. Overall, this project demonstrates an effective integration of frontend and backend technologies to deliver a functional, secure, and responsive job portal system for real world application.

# Thank You

## ANY QUERIES???